Question	Answer	Marks
3(a)	**One** mark per mark point (**Max 5**) **One** mark per mark point for **purpose** of Application Layer (**Max 3**) MP1 To provide services / interface with the user // access to applications, for example login, file transfer, network file access, email, etc MP2 To provide mechanisms for securing communication e.g. encryption/authentication MP3 To define/provide protocols used to allow the exchange of data/communication // to contain programs that exchange data MP4 Error detection and recovery mechanisms to handle application specific errors **One** mark per mark point for **purpose** of Transport Layer (**Max 3**) MP5 To provide logical communication between applications running on different hosts // To ensure that data is delivered to the correct application process on the destination machine MP6 To ensure error-free, end-to-end delivery of data between a source and a destination, in sequence // to provide error recovery techniques such as error detection codes and automatic repeat request MP7 To break data into segments when sent and to reconstruct when received // To reassemble segments at destination MP8 To regulate network connections // to provide flow control mechanisms to prevent data loss.	**5**
3(b)	**One** mark per mark point (**Max 4**) MP1 Data are broken into equal sized packets MP2 Data packets have headers containing information such as the IP addresses of the sender and receiver MP3 Each packet of data is sent independently to the destination // Packets don't necessarily follow the same route MP4 Each packet is sent via the most optimum path available MP5 Packets don't necessarily arrive in the order they were sent // Packets are reconstructed at the destination in the correct order MP6 Missing / damaged packets are re-sent	**4**